

human cell is composed of about 10^{14} atoms and the pancreas you want has at least 80 billion cells, probably more. We could use a scanning tunneling microscope to position individual atoms with some precision, but to make a macroscopic object with it would take a very long time.

The theoretical solution, initially, was exponential manufacturing, which we have discussed earlier; that nanobots could construct objects on its own scale. For instance, it could make another assembler, and each assembler could go on to make more assemblers, resulting in a suite of assemblers that would combine forces to make a macroscopic object.

According to Jones this is a seductive idea, seemingly validated by the workings of the cells of our own bodies. As he points out the human body is composed of nothing but a host of sophisticated nanoassemblers: molecular motors that convert chemical energy into mechanical energy and membranes with active ion channels that sort molecules--two key tasks needed for basic nanoscale assembly. Cell biology also exhibits software-controlled manufacturing, in the form of protein synthesis. The ribosome is a remarkable molecular machine that can read information from a strand of messenger RNA and convert the code into a sequence of amino acids. The ribosome fulfils the functions expected of an artificial assembler--proof that complex nanoassembly is possible.

So if biology can produce a sophisticated nanotechnology based on soft materials like proteins and lipids, singularitarian thinking goes, then how much more powerful our synthetic nanotechnology would be if we could use strong, stiff materials, like diamond. And if biology can produce working motors and assemblers using just the random selections of Darwinian evolution, how much more powerful the devices could be if they were rationally designed using all the insights we've learned from macroscopic engineering.

But that reasoning fails to take into account the physical environment in which cell biology takes place, which has nothing in common with the macroscopic world of bridges, engines, and transmissions. As Jones point out, in the domain of the cell,

- ▶ Water behaves like thick molasses, not the free-flowing liquid that we are familiar with.
- ▶ The fluctuations of constant Brownian motion ceaselessly bombarded components by fast-moving water molecules and flex and stretch randomly.
- ▶ The van der Waals force, which attracts molecules to one another, dominates, causing things in close proximity to stick together. Clingiest of all are protein molecules, whose stickiness underlies a number of undesirable phenomena, such as the rejection of medical implants.